Does Federal Reserve Communication Improve Policy-Rate Forecasts?

Evidence from Look-Ahead-Controlled, Real-Time Walk-Forward Tests

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Abstract

We ask whether the hawkish–dovish stance of Federal Reserve documents improves out-of-sample forecasts of the federal funds rate beyond macroeconomic statistics and market prices. Stance is measured with two readers: a chronologically consistent ChronoBERT classifier, retrained each year on earlier labels only, and sentence labels from the large language model Claude Haiku. We forecast the monthly average effective rate (308 test months) and the decision at each FOMC meeting (212 meetings) from 2001 to 2026 in expanding-window walk-forward tests, with ridge regression, a penalised ordered logit, and LSTM, GRU and Temporal Fusion Transformer networks. Adding text lowers squared error in most comparisons, but few gains survive testing. On a corrected monthly information cutoff, text lowers the root mean squared error of a statistics-only ridge model from 0.192 to 0.168 percentage points (one-sided Diebold–Mariano–HLN $p = 0.103$), and 78% of that gain comes from June 2020. Of six network comparisons per task, only an LSTM reading Haiku stance on meeting decisions survives Holm adjustment (adjusted $p = 0.020$); its gain is not statistically established with strictly real-time inputs, and Haiku's training data cover the whole test period. Once market prices are in the model, text changes monthly RMSE by less than 0.002 percentage points. Fed communication features improved forecasts in some specifications, but the evidence does not yet establish a consistent and reliable incremental benefit.

Keywords: central bank communication; monetary policy; text as data; large language models; look-ahead bias; forecast evaluation. **JEL:** E52, E58, C53.

*[Affiliation]. Working paper prepared for discussion with my supervisor. Code, result files and the audit trail are in the project repository. All results labelled *exploratory* were added after earlier results had been seen; every change to the pre-registered plans is dated in `DEVIATIONS.md`.

1 Introduction

The Federal Reserve communicates constantly. Post-meeting statements, minutes, press conferences and speeches are intended to shape expectations of the policy path, and asset prices respond to them (Gürkaynak et al., 2005; Hansen and McMahon, 2016). If this communication carries information about future decisions that is not already in data and prices, a forecaster who reads it should outperform one who does not. This paper tests that proposition under conditions designed to make a false positive difficult.

The question is deliberately narrow. Does the hawkish–dovish tone of Fed documents improve *out-of-sample* forecasts of the federal funds rate, *beyond* macroeconomic statistics and market prices that were available at the same moment? We study two tasks over 2001–2026: the monthly average effective federal funds rate and the target rate set at each meeting of the Federal Open Market Committee (FOMC).

Recent work reports that neural networks and large language models (LLMs) forecast the policy rate well (Chan-Lau et al., 2025). Two problems make such results hard to interpret. First, LLMs trained on recent text may have read about the very decisions they are asked to forecast (Lopez-Lira et al., 2025; Sarkar and Vafa, 2024; Crane et al., 2025). Second, macroeconomic data are revised, and back-tests that use today's vintages give models information that forecasters did not have (Croushore and Stark, 2001). A text gain measured without controlling for both can reflect hindsight rather than skill.

Our design addresses these problems in three ways.

1. **A chronologically consistent reader.** Our primary text reader is ChronoBERT (He et al., forthcoming), which releases one model per year trained only on text up to that year. We fine-tune each year's classifier only on labelled sentences from documents dated before that year. A second reader, Claude Haiku, is used for comparison, and its memory of the test period is measured directly.

2. **Real-time inputs.** Statistics are taken from ALFRED vintages released before each forecast cutoff. Results are reported separately for mixed-vintage, partially real-time and strictly real-time input sets.

3. **Pre-specification and controls.** Hypotheses, tests and multiple-testing families were fixed before the main results; later changes are dated and labelled. Every text model is paired with a placebo in which the text features are permuted or shifted.

The paper makes three contributions. It provides a pre-specified, look-ahead-controlled estimate of what Fed text adds to policy-rate forecasts across linear models and three neural architectures. It compares a domain-trained, time-consistent classifier with a general-purpose LLM on identical documents and protocol. And it documents how fragile a headline gain can be: to a single month, to the information cutoff, and to the composition of the multiple-testing family. An internal audit corrected two errors in our own earlier reporting, and the corrected numbers are used throughout.

The answer is modest. Text lowers forecast error more often than it raises it, and one network comparison survives correction. But the monthly gain rests largely on one month at the zero lower bound, the surviving meeting gain is not established with strictly real-time inputs and comes from the reader that has seen the test period, and market prices leave text almost nothing to add.

Section 2 reviews related work. Section 3 describes the data and Section 4 the methods. Section 5 presents the results and Section 6 the robustness analysis. Section 7 discusses limitations and Section 8 concludes.

2 Related literature

Central bank communication and asset prices. Gürkaynak et al. (2005) show that FOMC statements move asset prices through a "path" factor distinct from the surprise in the target rate, and Gürkaynak et al. (2007) show that market-based measures are strong predictors of the policy path. Lucca and Trebbi (2009) and Hansen and McMahon (2016) measure the content of Fed language directly. These studies mostly ask whether words move prices on announcement days. We ask a different question: whether words help forecast later decisions once prices are already known.

Text-based stance measures. Dictionary approaches (Loughran and McDonald, 2011) have given way to supervised classifiers trained on hand-labelled sentences. Trillion Dollar Words (Shah et al., 2023) and the World Central Banks dataset (Shah et al., 2025) provide hawkish, dovish and neutral labels for Fed sentences, and we use both. Newspaper sentiment (Shapiro et al., 2022), geopolitical risk (Caldara and Iacoviello, 2022) and economic policy uncertainty (Baker et al., 2016) provide further text-derived indicators.

Machine-learning forecasts of the policy rate. Chan-Lau et al. (2025) report accurate federal funds rate forecasts from LSTM and GRU networks and from an LLM. We attempted to replicate their LSTM and GRU results and did not succeed: on their January 2016 to April 2025 window, our real-time versions reach RMSE 0.415 and 0.647 percentage points against 0.238 and 0.271 published, while a no-change forecast scores 0.192.

Look-ahead bias in language models. Lopez-Lira et al. (2025) show that GPT-4o recalls economic series before its training cutoff and that masking and instructions do not remove this; Sarkar and Vafa (2024) and Crane et al. (2025) document related forms of look-ahead. The most common remedy is a chronologically consistent model (He et al., forthcoming). Glasserman and Lin (2023) and Engelberg et al. (2025) propose masking diagnostics, and Eliseev and Seleznev (2026) propose fake-date tests.

Forecast evaluation. We compare nested forecasts with the test of Diebold and Mariano (1995) and the small-sample correction of Harvey et al. (1997), Newey and West (1987) standard errors,

McNemar (1947) tests for decision accuracy and Holm (1979) adjustment within pre-registered families. The random walk is the standard benchmark (Meese and Rogoff, 1983).

3 Data

The sample runs from January 1994 to September 2026. Rows before 2001 are used for training only, and forecasts are evaluated from January 2001.

3.1 Targets

Table 1: Forecasting tasks

Task	Target	Test sample
T1, monthly	Monthly average effective federal funds rate (FRED `FEDFUNDS`); models forecast the change from the last observed month	308 months, Jan 2001–Sep 2026 (Jan 2004 dropped for missing data)
T2, per meeting	Target set at each FOMC meeting (midpoint of the range) and its class: cut, hold or hike, with a ±0.125 percentage-point band	212 meetings incl. 6 unscheduled; the network study uses the 206 scheduled meetings

3.2 Inputs

Table 2 lists the input blocks. Statistics (S) follow Chan-Lau et al. (2025) without the commercial property index, which has no free equivalent. Market variables (M) are FRED daily series as of the cutoff.

Table 2: Input blocks

Block	Content	Source
S, statistics (16)	Lagged policy rate; M2 (monthly and 12-month change); PCE and core PCE inflation; payroll change; average weekly earnings; nominal GDP growth; U-6 (monthly and 12-month change); PPI; CPI and core CPI; FHFA house prices; 10y–2y and 1y–6m yield slopes	FRED/ALFRED vintages
M, markets (8)	3-month bill and 6-month, 1-year and 2-year yield spreads to the target; target level; zero-lower-bound flag; VIX; WTI oil	FRED daily
N, news (6)	Fed stance and its change; Beige Book net tone; newspaper sentiment; geopolitical risk; policy uncertainty	Section 4.1
Fed text	2,840 documents: 240 statements, 263 minutes, 94 press-conference transcripts, 262 Beige Books, 1,981 Board speeches	federalreserve.gov
Labels	Hawkish, dovish and neutral sentences from Trillion Dollar Words and the Fed subset of World Central Banks	Shah et al. (2023, 2025)

3.3 Information cutoffs and data vintages

Each statistic is the latest ALFRED vintage released strictly before the forecast cutoff. Where ALFRED has no vintage, the earliest available vintage with a conservative release lag is used; those values include later revisions. First vintages begin in 2011 for average weekly earnings and U-6 and in 2012 for house prices. As a result, 136 of 308 monthly and 96 of 212 meeting test rows (44–45%) use at least one revised value. We therefore report three vintage treatments:

- **Mixed vintage**: the full input set, with disclosed fallback values;
- **Partial**: the four fallback columns removed from the inputs, although fallback values remain in pre-2001 training rows;
- **Strict real-time**: only the eight series with complete vintages (policy-rate lag, M2 monthly and yearly change, payrolls, nominal GDP, CPI and the two yield slopes), plus M for meetings. No fallback value enters tuning, training or prediction.

Monthly cutoff correction. The original monthly panel dated each forecast at the first calendar day of month t, when FEDFUNDS$(t-1)$ is not guaranteed to have been published. A corrected panel places the cutoff at 00:00 New York time at the start of the second US business day of month t. The network study used the corrected panel from the start; the linear monthly models were rerun on it after the audit (Section 6.1).

Label dating. The published year field of Trillion Dollar Words does not always match the document date, which would leak future labels into training. Each label is dated by the earliest corpus document that contains it; the 10.7% of Trillion Dollar Words and 1.7% of World Central Banks sentences that match no document are dropped.

4 Methods

4.1 Measuring stance

ChronoBERT (C). For each classifier year $y \geq 2000$ we start from the ChronoBERT backbone trained on text published up to 31 December of year $y-1$ (2025 and 2026 use the 2024 vintage). A linear layer on the mean-pooled last hidden state classifies each sentence as dovish, hawkish or neutral. It is trained only on labels whose source document is dated before 1 January of year y, with AdamW (learning rate 2×10^{-5}, weight decay 0.01), 3 epochs, batch size 16, at most 256 tokens, and three seeds whose probabilities are averaged. Documents of year y are scored only by the classifier of year y. Held-out sentence accuracy rises from 0.56 in 2000 to 0.70 in 2025 (mean 0.644), against 0.42–0.56 for always predicting the most common class.

Claude Haiku (H). The pinned model snapshot `claude-haiku-5-5` labels 343,742 sentences from 2,578 statements, minutes, press conferences and speeches. Sentences are masked, shuffled and sent 25 per request; 343,717 receive a label, of which 8.5% are hawkish and 12.4% dovish. On a pilot sample Haiku agrees with the human labels on 67.3% of sentences. Its training data extend to June 2026, so it has seen the entire test window (Section 7).

Aggregation. For sentence i the stance score is

$$s_i = \begin{cases} \hat{P}_i(\text{hawkish}) - \hat{P}_i(\text{dovish}) & \text{ChronoBERT,} \\ +1,\ 0,\ -1 \text{ for hawkish, neutral, dovish} & \text{Haiku.} \end{cases} \tag{1}$$

A document's stance is the mean of s_i over its sentences of 30–1,000 characters. For each forecast row, the stance is the mean of three group means—statements with press conferences, minutes, and speeches—computed over documents released since the previous cutoff and carried forward when no document appears:

$$\text{stance}_t = \tfrac{1}{|G_t|} \sum_{g \in G_t} \ \tfrac{1}{|D_{g,t}|} \sum_{d \in D_{g,t}} \bar{s}_d, \tag{2}$$

where G_t is the set of non-empty groups. The text block used in the network and Study 4 ordered-logit models (C or H) is stance_t and its change from the previous row. Block N in the Plan 1 linear models adds Beige Book tone (Loughran–McDonald net tone), newspaper sentiment, and the logs of the geopolitical-risk and policy-uncertainty indices.

4.2 Forecasting models

Benchmarks. The no-change forecast sets next month's rate to the last observed value and holds the target at each meeting. An inertial Taylor rule (Taylor, 1993) regresses the rate on its lag, real-time core PCE inflation and real-time U-3 unemployment.

Ridge regression (T1). The monthly model forecasts the change in the rate,

$$\hat{\beta} = \arg\min_{\beta} \sum_{\tau \le t-2} \left(\Delta r_\tau - x_\tau' \beta\right)^2 + \lambda \|\beta\|_2^2, \qquad \hat{r}_t = r_{t-1} + x_t' \hat{\beta}, \tag{3}$$

with λ chosen from seven log-spaced values between 0.1 and 1,000.

Penalised ordered logit (T2). For the decision class $y_m \in \{\text{cut} < \text{hold} < \text{hike}\}$ at meeting m,

$$\Pr(y_m \le j \mid x_m) = \Lambda(\kappa_j - x_m' \beta), \qquad j \in \{\text{cut}, \text{hold}\}, \tag{4}$$

estimated with an L2 penalty on β chosen from $\{0.01, 0.03, 0.1, 0.3, 1, 3\}$. The class forecast is the most probable class; the rate forecast is the standing target plus the probability-weighted mean move, with mean move sizes per class taken from the training window.

Neural networks. We use an LSTM (Hochreiter and Schmidhuber, 1997), a GRU (Cho et al., 2014) and a single-horizon network inspired by the Temporal Fusion Transformer (TFT; Lim et al., 2021), comprising variable-selection networks, an LSTM encoder, gated skip connections and interpretable multi-head attention. All networks forecast the change in the rate, are trained with Adam on mean squared error, and average eight seeds. Hyperparameters are tuned inside each training window on five identical validation folds per candidate, scored in rate units, and the networks are refit every 12 months or eight meetings (the version 2 protocol, Section 4.6).

4.3 Walk-forward design

Every model is re-estimated on an expanding window and forecasts only rows it has never seen. The forecast for month t is trained on rows up to $t-2$, because the outcome for month $t-1$ is not known at the cutoff. Meeting forecasts are trained on earlier meetings only. Standardisation, penalty selection, hyperparameter search and early stopping use training rows only; penalties are chosen by five-fold expanding-window cross-validation.

4.4 Controls

The *placebo* replaces the text features with uninformative versions: permuted across rows inside each training window in the linear models, and circularly shifted by half the panel in the networks. A placebo gain would signal that text helps through spurious structure. The *shuffled-label* control permutes training targets, to check that the pipeline cannot forecast without signal.

4.5 Tests

Let $e_{b,t}$ and $e_{x,t}$ be the forecast errors of the base model and the model with text, and define the loss differential

$$d_t = e_{b,t}^2 - e_{x,t}^2, \qquad \Delta\text{MSE} = \bar{d} = \tfrac{1}{n}\textstyle\sum_t d_t, \tag{5}$$

so that $\Delta\text{MSE} > 0$ favours text. The registered test for the linear models is the one-sided Diebold–Mariano statistic with the Harvey–Leybourne–Newbold correction,

$$\text{DM-HLN} = \sqrt{\frac{n+1-2h+h(h-1)/n}{n}}\;\frac{\bar{d}}{\sqrt{\hat{V}(\bar{d})}} \;\sim\; t_{n-1}, \tag{6}$$

with horizon $h = 1$. The network test is a one-sided t-test on $\bar{d}$ with Newey and West (1987) standard errors, using lags equal to the refit interval (12 months or 8 meetings). Effect sizes are reported as ΔMSE with a 90% paired moving-block bootstrap interval (block length equal to the refit interval, 10,000 draws, seed 0). Decision accuracy is compared with McNemar's exact test on the meetings one model fixes and the other breaks.

Within a family of m tests with ordered raw p-values $p_{(1)} \le \cdots \le p_{(m)}$, Holm-adjusted values

are

$$\tilde{p}_{(k)} = \max_{j \le k} \; \min\{1, \; (m - j + 1)\, p_{(j)}\}. \tag{7}$$

4.6 Pre-specification and analysis roles

Table 3 lists the testing families. Plan 1 (linear models) and Plan 2 (first neural models) were registered with git tags before results. The version 2 network protocol was written on 13 October 2026, after the version 1 network results had been seen and an external review had found defects; it is a dated amendment, not a pre-registration, and version 1 results are superseded. Analyses added after results are labelled exploratory.

Table 3: Testing families and outcomes

Study	Family	Size	Survivors at 0.05
Plan 1, H1 (registered)	Conjunctive: monthly DM-HLN, meeting DM-HLN and McNemar must all pass	3	None
Plans 1–2, H2 (registered)	Holm across the H2 tests that ran (H2a 10, H2c 2, H2d 3)	15	1, H2c (not a text increment)[a]
Study 4, ordered logit	Single primary	1	None ($p = 0.144$)
Study 4 networks, T2 (amendment)	Holm, 3 architectures × 2 readers	6	LSTM + Haiku (0.020)
Study 4 networks, T1 (amendment)	Holm, 3 architectures × 2 readers	6	None (min. 0.156)
Study 5, three meetings ahead	Single primary; Holm across 4 secondaries	1+4	None

[a] H2c compares the meeting model with text against the random walk ($p = 0.0044$); it does not measure what text adds over a model without text.

5 Results

Throughout, RMSE is in percentage points and ΔMSE in squared percentage points, with positive values favouring text. Figure 1 shows the monthly forecasts against the actual rate.

5.1 Linear models

No registered linear hypothesis is supported (Table 4). Adding text to statistics gives the largest relative gain in the monthly task, a 13% lower RMSE, but its 90% interval includes zero and the registered test gives $p = 0.103$. Once market prices are included, text changes monthly RMSE by 0.0017 and meeting RMSE by 0.0002 percentage points.

Decision accuracy tells the same story. With markets in the meeting model, accuracy rises from 86.8% to 91.0% when the full news block is added (11 meetings fixed, 2 broken). The raw McNemar p is 0.011, but 0.146 after Holm adjustment across the 15-test H2 family. Haiku stance raises accuracy from 86.8% to 89.2% (6 fixed, 1 broken; McNemar $p = 0.062$).

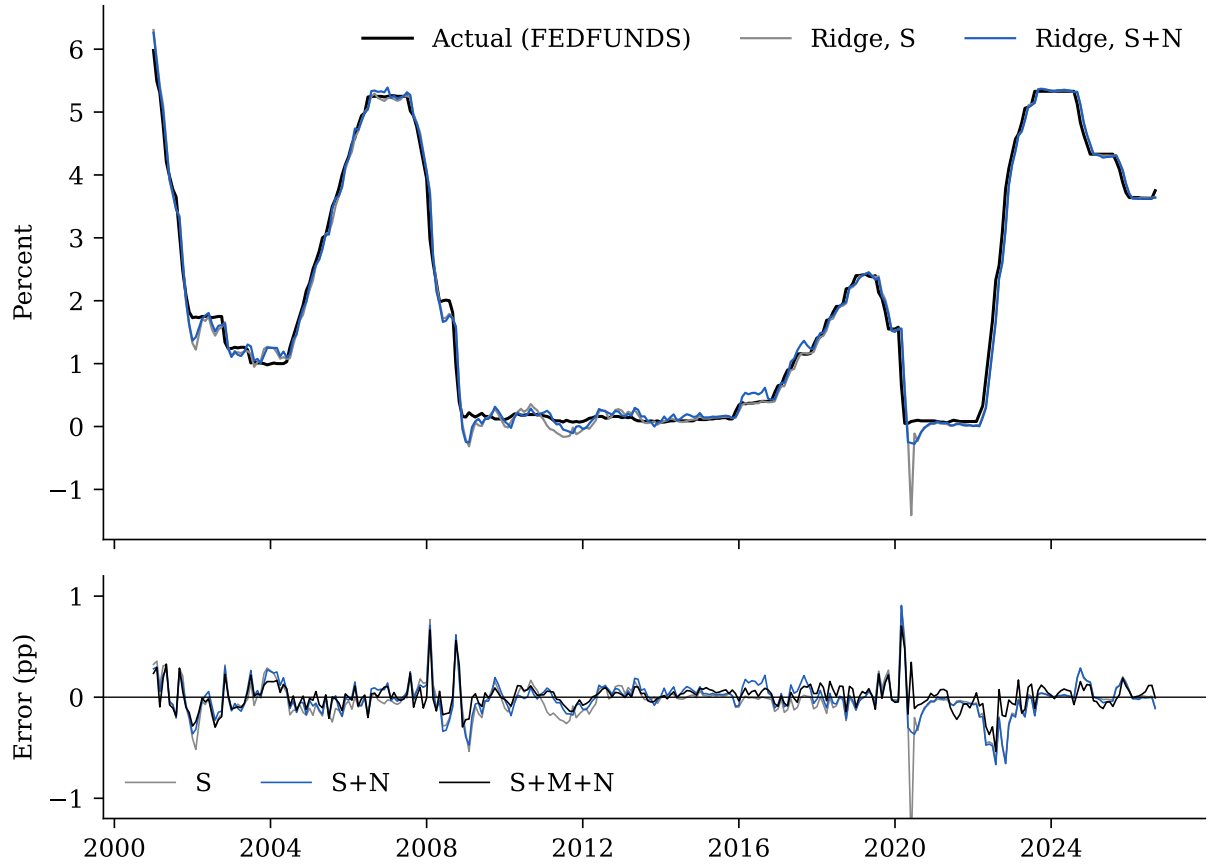


Figure 1: Monthly federal funds rate and walk-forward ridge forecasts on the corrected cutoff, January 2001 to September 2026. Lower panel: forecast errors (forecast minus actual) for the statistics-only (S), statistics plus news (S+N) and full (S+M+N) models.

Text also fails to help at longer horizons. In Study 5, which uses strictly real-time inputs, adding ChronoBERT stance to forecasts three meetings ahead raises log loss from 0.779 to 0.839 ($p = 0.898$, 204 meetings).

5.2 Neural networks

Table 5 and Figure 2 report the twelve registered network comparisons. Ten of the twelve point estimates favour text, but only one survives Holm adjustment. On scheduled meetings, the LSTM with Haiku stance lowers RMSE from 0.1420 to 0.1333, $\Delta\text{MSE} = +0.0024$ [+0.0010, +0.0039], Newey–West $p = 0.0034$, adjusted $p = 0.020$. Its decision accuracy rises from 78.6% to 84.5% (15 meetings fixed, 3 broken). The same LSTM with ChronoBERT stance gains far less (adjusted $p = 0.85$), so the result depends on the reader.

In the monthly task three text arms have raw p-values below 0.05 (LSTM + Haiku 0.026, TFT + Haiku 0.034, TFT + ChronoBERT 0.042) and bootstrap intervals above zero, but none survives adjustment; the smallest adjusted p is 0.156.

Table 4: Linear models: forecast gains from adding text

Model and task	Comparison	Without text	With text	ΔMSE [90% CI]	p
Ridge, T1 (308 months)[a]	S → S+N	0.1919	0.1675	+0.0088 [−0.0001, +0.0219]	0.103
Ordered logit, T2 (212)	S → S+N	0.2052	0.1994	+0.0023 [−0.0018, +0.0048]	0.061
Ridge, T1 (308)[a]	S+M → S+M+N	0.1333	0.1316	+0.0004 [−0.0011, +0.0025]	0.251
Ordered logit, T2 (212)	S+M → S+M+N	0.1321	0.1323	−0.0000 [−0.0012, +0.0009]	0.520
Study 4, log loss					
Ordered logit, T2 (212)	S+M → S+M+H	0.2938	0.2721	+0.0217 [−0.0085, +0.0611]	0.144
Ordered logit, T2 (212)	S+M → S+M+C	0.2938	0.2814	+0.0123 [−0.0079, +0.0391]	0.207

Columns 3–4: RMSE (pp), or log loss in the Study 4 rows. p: one-sided DM-HLN on squared error (log loss in Study 4). Intervals: paired moving-block bootstrap. S: statistics; M: markets; N: Plan 1 news block; H, C: Haiku and ChronoBERT stance.

[a] Corrected monthly cutoff (post-audit rerun, Section 6.1).

Table 5: Neural networks, mixed-vintage inputs: registered comparisons

Task	Network + reader	Base	+ text	ΔMSE [90% CI]	NW p	Holm p	Accuracy (%)
T2	LSTM + Haiku	0.1420	0.1333	+0.0024 [+0.0010, +0.0039]	0.003	**0.020**	78.6 → 84.5
T2	LSTM + ChronoBERT	0.1420	0.1406	+0.0004 [−0.0012, +0.0028]	0.375	0.85	78.6 → 83.5
T2	GRU + Haiku	0.1445	0.1379	+0.0019 [−0.0007, +0.0063]	0.200	0.85	80.6 → 82.0
T2	GRU + ChronoBERT	0.1445	0.1612	−0.0051 [−0.0089, −0.0019]	0.992	0.99	80.6 → 80.1
T2	TFT + Haiku	0.1415	0.1342	+0.0020 [−0.0008, +0.0061]	0.170	0.85	86.4 → 86.9
T2	TFT + ChronoBERT	0.1415	0.1361	+0.0015 [−0.0009, +0.0051]	0.210	0.85	86.4 → 85.9
T1	LSTM + Haiku	0.1878	0.1722	+0.0056 [+0.0007, +0.0089]	0.026	0.156	59.1 → 65.3
T1	LSTM + ChronoBERT	0.1878	0.1979	−0.0039 [−0.0100, +0.0007]	0.878	1.00	59.1 → 53.9
T1	GRU + Haiku	0.1803	0.1779	+0.0008 [−0.0105, +0.0089]	0.447	1.00	60.4 → 59.1
T1	GRU + ChronoBERT	0.1803	0.1782	+0.0008 [−0.0027, +0.0042]	0.360	1.00	60.4 → 62.3
T1	TFT + Haiku	0.1911	0.1726	+0.0067 [+0.0018, +0.0137]	0.034	0.168	54.9 → 59.4
T1	TFT + ChronoBERT	0.1911	0.1766	+0.0053 [+0.0008, +0.0107]	0.042	0.169	54.9 → 56.5

T2: 206 scheduled meetings, base S+M. T1: 308 months on the corrected cutoff, base S. Base and +text: RMSE (pp). NW p: one-sided Newey–West test on ΔMSE. Holm over the six comparisons within each task. Accuracy: meeting decision class (T2) or direction of the monthly average-rate change (T1). TFT: the single-horizon TFT-inspired network.

5.3 Which model forecasts best

Market prices dominate. The best monthly model is ridge with S+M+N (RMSE 0.1316), nearly tied with S+M (0.1333) and well ahead of the no-change forecast (0.1813). On meetings, the ordered logit with S+M+N has the highest accuracy, 90.8% on the 206 scheduled meetings, while the lowest meeting RMSE (0.122) belongs to the strict-input GRU without text. These rankings were not tested and are descriptive. The linear models match or beat the networks at a small fraction of the computational cost.

6 Robustness

Sections 6.1 and 6.3 were added after the main results had been seen and are exploratory.

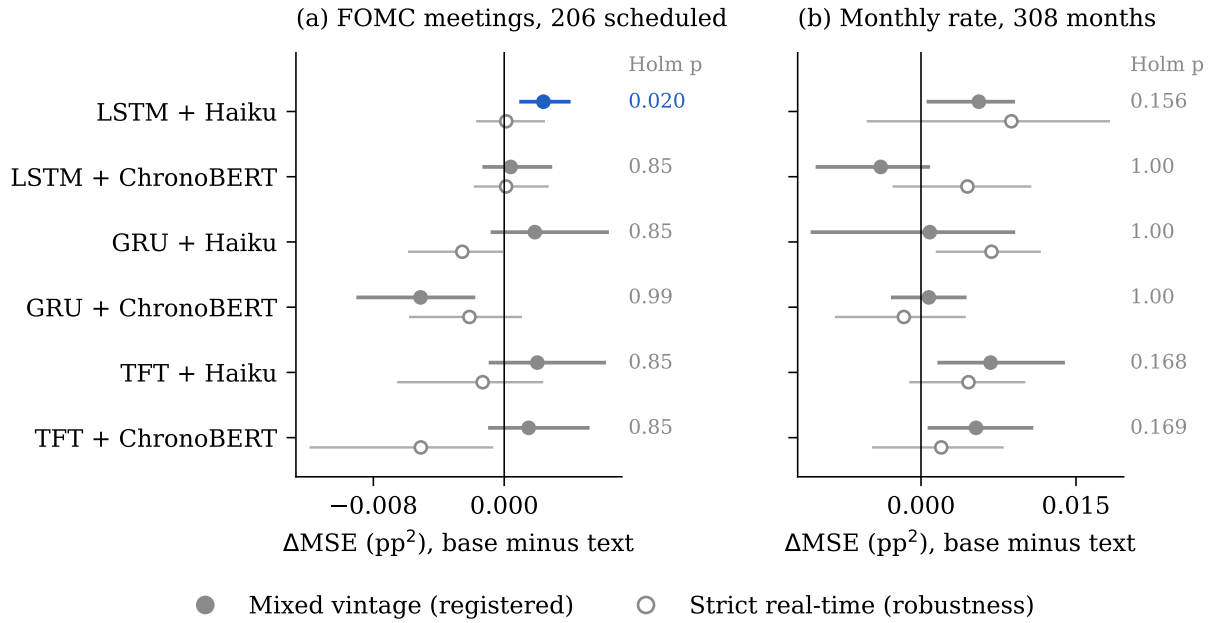


Figure 2: Gain from adding Fed stance to the networks: ΔMSE (base minus text, pp^2) with 90% paired block-bootstrap intervals. Filled markers: registered mixed-vintage comparisons, with Holm-adjusted p on the right. Hollow markers: the same comparisons with strictly real-time inputs (robustness, outside the Holm family). Bootstrap intervals are unadjusted.

6.1 Corrected monthly cutoff

Table 6 reruns the Plan 1 monthly models on the corrected cutoff with all settings unchanged. The statistics-only model loses information it should not have had, and its RMSE rises from 0.1798 to 0.1919; the model with text barely moves. The registered test for the headline comparison moves from $p = 0.077$ to 0.103. The meeting models and networks already used valid cutoffs and are unaffected.

Table 6: Plan 1 monthly ridge models on the original and corrected cutoffs

	RMSE (pp)		ΔMSE [90% CI]	p	
Comparison	Corrected	Original	Corrected	Corrected	Original
S → S+N (H1)	0.1919 → 0.1675	0.1798 → 0.1664	+0.0088 [−0.0001, +0.0219]	0.103	0.077
S → placebo	0.1919 → 0.1917	0.1798 → 0.1890	+0.0001 [−0.0005, +0.0012]	0.439	0.858
No change → shuffled	0.1813 → 0.1824	0.1813 → 0.1823	−0.0004 [−0.0020, +0.0013]	0.750	0.737
No change → S+N (H2c)	0.1813 → 0.1675	0.1813 → 0.1664	+0.0048 [−0.0042, +0.0107]	0.074	0.056
S+M → S+M+N (H2d)	0.1333 → 0.1316	0.1311 → 0.1332	+0.0004 [−0.0011, +0.0025]	0.251	0.695

308 test months. RMSE in pp. p: one-sided DM-HLN on squared error.

6.2 Strictly real-time inputs

With only fully vintaged series, the LSTM + Haiku meeting gain falls to ΔMSE = +0.0001 [−0.0016, +0.0024] (RMSE 0.1251 to 0.1246, $p = 0.461$; hollow markers in Figure 2). The improvement was

not statistically established in the strict-input analysis. A significant result beside a non-significant one does not by itself show that the two differ, and because the strict and mixed models use different inputs, their RMSE levels are compared only descriptively. Of the twelve strict comparisons one has a raw p below 0.05, the monthly GRU + Haiku (ΔMSE = +0.0068 [+0.0015, +0.0115], $p = 0.014$); strict arms lie outside the Holm family.

6.3 Influence of June 2020

Figure 3 plots the running sum of d_t for the monthly S $\rightarrow$ S+N comparison. June 2020 alone supplies 78% of the total gain. Without that month ΔMSE falls from +0.0088 to +0.0019 and RMSE moves only from 0.1723 to 0.1666; on the original cutoff the share is 67%. The monthly text gain is therefore mostly a single forecast at the zero lower bound rather than a steady improvement.

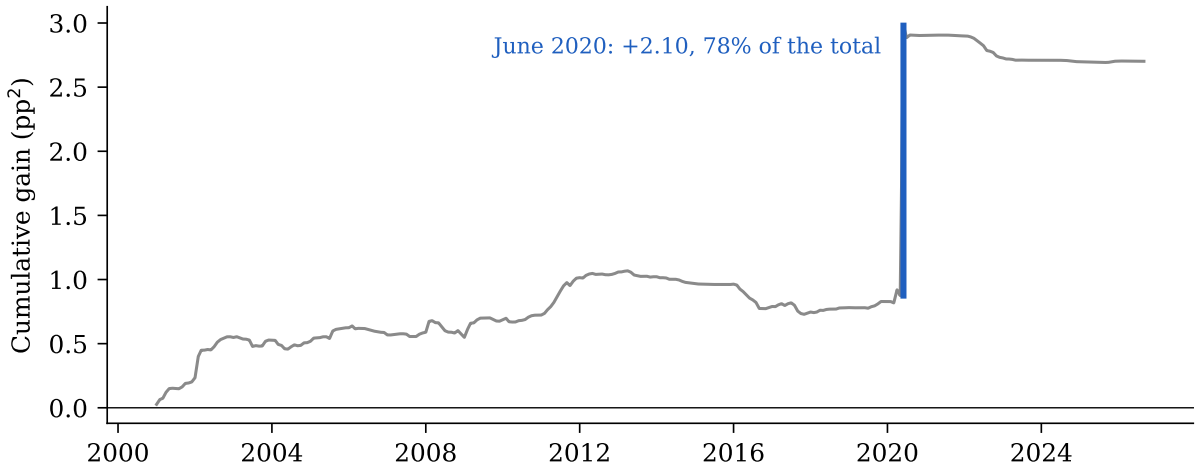


Figure 3: Running sum of the squared-error gain of ridge S+N over ridge S (pp^2), corrected monthly cutoff, 308 months. A rising line means text is helping.

6.4 Placebo controls

No placebo arm improves significantly on its base. In the linear monthly model the placebo moves RMSE from 0.1919 to 0.1917 ($p = 0.439$), and the Study 4 Haiku placebo gives $p = 0.441$. Across the twelve network placebo arms the smallest one-sided p against the base model is 0.115. The LSTM + Haiku meeting model also beats its own placebo (ΔMSE = +0.0031 [+0.0014, +0.0046], $p = 0.0008$). Clean placebos argue against spurious structure in the pipeline, but they cannot detect a reader that recalls outcomes.

6.5 Interim versus final adjustments

Two figures circulated in earlier drafts came from incomplete families and are superseded. An adjusted p of 0.014 for LSTM + Haiku was a Holm adjustment over four tests, computed before

the TFT arms had finished; over the registered six it is 0.020. An adjusted McNemar p of 0.045 for the meeting accuracy gain used an interim family; over the final 15 H2 tests it is 0.146. The code now refuses to report an adjustment until every member of the registered family is present.

7 Discussion and limitations

The surviving result is the one most exposed to look-ahead. Haiku's training data run to June 2026, beyond every test meeting. Asked for decisions by date alone, it recalls 87.8% of them, and it dates 89.5% of masked statements to within one year. A regression of Haiku's gain over ChronoBERT on whether Haiku recalled the decision finds nothing ($p = 0.378$), but only 21 of 211 meetings were not recalled, so that test has little power (see Lopez-Lira et al., 2025, on why memorisation evidence is a lower bound). The LSTM + Haiku gain therefore cannot be read as skill a forecaster would have had in real time. ChronoBERT, the look-ahead-controlled reader, yields no gain that survives adjustment.

Further limitations qualify the results.

- **Power.** 308 months and 212 meetings are few for gains of a few percent of RMSE, and errors within a policy cycle are correlated. No minimum detectable effect was registered, so a non-significant result is not evidence of no effect.
- **Zero lower bound.** About a third of the test window lies at the zero lower bound, where the no-change forecast is nearly perfect and a single month can dominate.
- **Revised data.** In the mixed-vintage results 44–45% of test rows contain a revised statistic. These values enter every model identically, but that does not guarantee the bias cancels between models with and without text.
- **Hindsight in labels and indices.** The human stance labels were written in 2023–2025, and the newspaper-sentiment, geopolitical-risk and policy-uncertainty indices were constructed later with fixed methods; their historical revisions could not be verified.
- **Classifier quality.** ChronoBERT is about 64% accurate on held-out sentences, and the sentence-level split makes that figure likely optimistic.
- **Protocol timing.** The network protocol was written after version 1 results were seen. It is disclosed and dated, but it is not a pre-registration.
- **Scope.** One country, one target and one stance score per document. Richer representations of Fed text were not tested.

8 Conclusion

Fed communication features improved forecasts in some specifications, but the evidence does not yet establish a consistent and reliable incremental benefit. Text lowers error in most comparisons,

yet only one registered test survives adjustment; that result uses a reader that has seen the test period and is not established with strictly real-time inputs. Market prices already carry most of what Fed text says about the next decision.

Three extensions would sharpen the answer. A longer or multi-country sample, or live forecasts made after a reader's training cutoff, would raise power and remove memorisation. Richer measures than a single stance score—for example the content of forward guidance, or the surprise relative to market expectations—may capture what a tone index misses. A complete real-time data set for the statistics would remove the remaining revised values. Until then, the safest reading is that Fed text is a weak and unstable complement to market prices, not a substitute for them.

References

Scott R. Baker, Nicholas Bloom, and Steven J. Davis. Measuring economic policy uncertainty. *Quarterly Journal of Economics*, 131(4):1593–1636, 2016.

Dario Caldara and Matteo Iacoviello. Measuring geopolitical risk. *American Economic Review*, 112 (4):1194–1225, 2022.

Jorge A. Chan-Lau, Toan Long Quach, and Allen R. Shi. Forecasting federal funds rates with AI: LSTM, GRU, and large language model approaches. AMRO Working Paper WP/25-10, ASEAN+3 Macroeconomic Research Office, 2025.

Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using RNN encoder–decoder for statistical machine translation. In *Proceedings of EMNLP 2014*, pages 1724–1734, 2014.

Leland Crane, Akhil Karra, and Paul Soto. Total recall? evaluating the macroeconomic knowledge of large language models. Finance and Economics Discussion Series 2025-044, Board of Governors of the Federal Reserve System, 2025.

Dean Croushore and Tom Stark. A real-time data set for macroeconomists. *Journal of Econometrics*, 105(1):111–130, 2001.

Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business and Economic Statistics*, 13(3):253–263, 1995.

Alexander Eliseev and Sergei Seleznev. Fake date tests: Can we trust in-sample accuracy of LLMs in macroeconomic forecasting? arXiv:2601.07992; Bank of Russia Working Paper 167, 2026.

Joseph Engelberg, Asaf Manela, William Mullins, and Luka Vulicevic. Entity neutering. SSRN Working Paper 5182756, 2025.

Paul Glasserman and Caden Lin. Assessing look-ahead bias in stock return predictions generated by GPT sentiment analysis. arXiv:2309.17322, 2023.

Refet S. Gürkaynak, Brian Sack, and Eric T. Swanson. Do actions speak louder than words? the response of asset prices to monetary policy actions and statements. *International Journal of Central Banking*, 1(1):55–93, 2005. Reference added from memory; verify before circulation.

Refet S. Gürkaynak, Brian Sack, and Eric T. Swanson. Market-based measures of monetary policy expectations. *Journal of Business and Economic Statistics*, 25(2):201–212, 2007.

Stephen Hansen and Michael McMahon. Shocking language: Understanding the macroeconomic effects of central bank communication. *Journal of International Economics*, 99:S114–S133, 2016. Reference added from memory; verify before circulation.

David Harvey, Stephen Leybourne, and Paul Newbold. Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2):281–291, 1997.

Songrun He, Linying Lv, Asaf Manela, and Jimmy Wu. Chronologically consistent large language models. *Journal of Financial Economics*, forthcoming. arXiv:2502.21206.

Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8): 1735–1780, 1997.

Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.

Bryan Lim, Sercan Ö. Arık, Nicolas Loeff, and Tomas Pfister. Temporal fusion transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37(4): 1748–1764, 2021.

Alejandro Lopez-Lira, Yuehua Tang, and Mingyin Zhu. The memorization problem: Can we trust LLMs' economic forecasts? arXiv:2504.14765, 2025.

Tim Loughran and Bill McDonald. When is a liability not a liability? textual analysis, dictionaries, and 10-Ks. *Journal of Finance*, 66(1):35–65, 2011.

David O. Lucca and Francesco Trebbi. Measuring central bank communication: An automated approach with application to FOMC statements. NBER Working Paper 15367, National Bureau of Economic Research, 2009. Reference added from memory; verify before circulation.

Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947.

Richard A. Meese and Kenneth Rogoff. Empirical exchange rate models of the seventies: Do they fit out of sample? *Journal of International Economics*, 14(1–2):3–24, 1983.

Whitney K. Newey and Kenneth D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987.

Suproteem K. Sarkar and Keyon Vafa. Lookahead bias in pretrained language models. Working paper, 2024.

Agam Shah, Suvan Paturi, and Sudheer Chava. Trillion dollar words: A new financial dataset, task and market analysis. In *Proceedings of ACL 2023*, pages 6664–6679, 2023.

Agam Shah, Siddhant Sukhani, Huzaifa Pardawala, et al. Words that unite the world: A unified framework for deciphering central bank communications globally. In *NeurIPS 2025 Datasets and Benchmarks Track*, 2025. arXiv:2505.17048.

Adam Hale Shapiro, Moritz Sudhof, and Daniel J. Wilson. Measuring news sentiment. *Journal of Econometrics*, 228(2):221–243, 2022.

John B. Taylor. Discretion versus policy rules in practice. *Carnegie-Rochester Conference Series on Public Policy*, 39:195–214, 1993.

A Strict real-time network results

Table 7: Neural networks, strictly real-time inputs (robustness)

Task	Network + reader	Base	+ text	ΔMSE [90% CI]	NW p	Accuracy (%)
T2	LSTM + Haiku	0.1251	0.1246	+0.0001 [−0.0016, +0.0024]	0.461	81.6 → 85.4
T2	LSTM + ChronoBERT	0.1251	0.1247	+0.0001 [−0.0018, +0.0026]	0.467	81.6 → 84.5
T2	GRU + Haiku	0.1222	0.1323	−0.0026 [−0.0058, −0.0001]	0.925	83.0 → 86.4
T2	GRU + ChronoBERT	0.1222	0.1306	−0.0021 [−0.0058, +0.0010]	0.849	83.0 → 85.9
T2	TFT + Haiku	0.1358	0.1405	−0.0013 [−0.0065, +0.0023]	0.688	87.4 → 87.9
T2	TFT + ChronoBERT	0.1358	0.1534	−0.0051 [−0.0118, −0.0007]	0.935	87.4 → 86.9
T1	LSTM + Haiku	0.1972	0.1736	+0.0087 [−0.0051, +0.0182]	0.117	53.6 → 58.4
T1	LSTM + ChronoBERT	0.1972	0.1854	+0.0045 [−0.0026, +0.0106]	0.132	53.6 → 55.5
T1	GRU + Haiku	0.1841	0.1645	+0.0068 [+0.0015, +0.0115]	0.014	59.4 → 63.6
T1	GRU + ChronoBERT	0.1841	0.1885	−0.0017 [−0.0082, +0.0042]	0.669	59.4 → 59.4
T1	TFT + Haiku	0.1859	0.1731	+0.0046 [−0.0010, +0.0100]	0.084	53.6 → 56.5
T1	TFT + ChronoBERT	0.1859	0.1806	+0.0020 [−0.0046, +0.0079]	0.305	53.6 → 50.0

Base: eight fully vintaged statistics (plus M in T2). Other definitions as in Table 5. Strict arms are robustness checks and are not Holm-adjusted.

B Provenance of results

Table 8: Where each result comes from

Result	Code	Commit, date	Status
Plan 1 linear tests, original cutoff	`walkforward.py`, `evaluate.py`	8c8d3c2, 2026-10-09, after tag `prereg`	Registered; monthly part superseded
Plan 1 monthly rerun, corrected cutoff	`audit/2026-10-13/rerun_plan1_t1b.py`	2026-10-13, not yet committed	Post-audit correction, exploratory
Study 4 ordered logit	`haiku_compare.py`	7879a70, 2026-10-12; design b4a00fd before the run	Primary of an exploratory study
Study 4 networks, version 2	`tune.py`	1a31c73, 2026-10-13; protocol 6048e0d	Amendment written after v1 results
Study 4 networks, version 1	`tune.py` at 74ab18f	74ab18f	Superseded, not cited
Study 5, three meetings ahead	`study5.py`	074d2b9, 2026-10-12; design 85de61f	Primary of an exploratory study

The full audit, with recomputed tables for each vintage treatment, is in `audit/2026-10-13/AUDIT.md`; figures are produced by `paper/make_figures.py` from the committed result files.